ENBC332: Homework #5

Due: Monday, 10/27/25 by noon, beginning of the class

Submit your homework to ELMS, including R code and dataset

Overall points: 40

Note: Late homework may be submitted up to 24 hours after the due date, but a 50% penalty will be applied.

Note: Please aim to write clean, well-organized code that is easy for others to understand. Be sure to include comments or explanations where needed. Points may be deducted if this is not done.

Note: Please name your codes using the following format: "problem1_fullname.R". Start with the problem number and followed by your full name, e.g., "**problem1_jimmyazarnoosh**".

Note: Using generative AI for learning purposes is allowed. However, directly copying or closely replicating AI-generated content is strictly prohibited and will result in a grade of **ZERO**. Repeated violations may lead to more severe consequences.

Note: Submit **a single zip file** containing the following:

- The dataset you used
- R code files
- Your answers to the questions in PDF format

Note: Functions are recommended to follow the format below:
e.g., calculating range.

```
# -----------------------------------------------------
Data <- trees$Height
my_range_func <- function(Data){
  Range <- max(Data) - min(Data)
  return(Range)
}
# -----------------------------------------------------
```

Confidence Interval

Problem 1. The goal of this problem is to understand and apply the concept of the confidence interval. Use the dataset you already used in Homework 2 and write a code to complete the following tasks. Make sure your dataset is large enough with at least 1000 sample data. If not, find another dataset online and share it with others on the discussion board. In the Excel file that contains your dataset, you can have only ONE column or feature derived from the original dataset. Choose a meaningful variable/feature from the original dataset such as age, weight, height, etc. (40 points)

a) **Population variance is KNOWN:** (10 points)

Collect a random sample of 500 from your dataset. Using the sample data, calculate 90%, 95%, and 99% confidence intervals for the mean in the population. Note that population variance is known, and you can calculate variance or standard deviation from your data. Make sure you are using a proper test, z-test or t-test.

b) **Population variance is UNKNOWN:** (20 points)

In this part, we collect sample data from the population and discard the original dataset to simulate a scenario where the full population data is unavailable. We only use the sample data here.

Collect a random sample of 10, 50, 100, 300, 500, and 1000 from your dataset. Using these sample data, calculate 90%, 95%, and 99% confidence intervals for the mean in the population. Note that the population variance is unknown, and you cannot calculate variance or standard deviation from the original dataset. Make sure you are using a proper test, z-test or t-test.

c) Discuss your observation from parts a and b. Interpret the confidence interval in the context of the study. What does it tell you about the population mean? In part b, **plot** the sample size versus the width of the 90%, 95%, and 99% confidence intervals for the mean and discuss your observation. (10 points)